HERITEDGE

From Generation to Generation

"Understanding the existing digital heritage ecosystem and identifying the gap HeritEdge addresses"

Team HeritEdge

SIH 2026

HeritEdge – Technology Stack

Frontend

React.js / Next.js
Modern web framework for building the interactive HeritEdge interface, including heritage contribution, discovery, multilingual records, AI-powered Q&A and community verification.

Backend

Python + FastAPI
High-performance backend responsible for API orchestration, heritage processing workflows, AI integration, database operations, semantic retrieval and community verification.

AI & Knowledge Layer

Whisper — Speech Recognition
Converts community-contributed audio and video into searchable text while preserving the original spoken language.

IndicTrans2 — Multilingual Translation
Enables translation across Indian languages while retaining the original-language record, allowing heritage knowledge to be discovered across linguistic boundaries.

GPT-6 Astra — LLM & Reasoning Layer
Converts unstructured cultural content into structured heritage information and generates grounded responses using retrieved evidence, while explicitly distinguishing documented, community-submitted and unverified information.

Neo4j — Cultural Knowledge Graph
Represents relationships between traditions, regions, languages, communities, practitioners, sources and cultural variations, enabling relationship-aware heritage discovery.

BGE-M3 — Multilingual Embeddings
Converts multilingual heritage records and user queries into semantic vector representations for cross-language similarity search and retrieval.

Vector Database
FAISS — MVP Semantic Retrieval
Qdrant / pgvector — Production-scale Vector Search

RAG — Evidence-Grounded Knowledge Retrieval
Retrieves relevant heritage records and supporting evidence before generating responses, reducing hallucination and allowing newly contributed knowledge to become searchable without retraining the LLM.

PostgreSQL

PostgreSQL — Core Data Layer
Stores structured heritage records, contributor information, metadata, source provenance, verification status and platform-level data.

OpenStreetMap + Leaflet

OpenStreetMap + Leaflet — Geospatial Discovery
Provides an interactive map for exploring heritage records by geographic region, enabling location-based discovery of cultural traditions.

Object Storage

Object Storage — Media Preservation
Stores original audio, video, images and supporting cultural documents separately from structured database records while maintaining their provenance and references.

Cost Structure

MVP / SIH Prototype

Component	Technology	Pricing model	MVP cost
Frontend	React / Next.js	Open source + free hosting option	₹0
Backend	Python + FastAPI	Open source + free hosting option	₹0
Speech recognition	Whisper	Open-source model	₹0
Translation	IndicTrans2	$10/M input + $50/M output tokens	₹0
LLM	GPT-6 Astra	Open-source/self-hosted	
Embeddings	BGE-M3	MIT open source	₹0
Vector search	FAISS	Software layer	₹0
RAG	Custom implementation	Free tier	₹0
Database	PostgreSQL / Supabase Free	Future/production layer	₹0
Knowledge graph	Neo4j	Open-source	₹0 for MVP
Maps	OpenStreetMap + Leaflet	1 GB file storage	₹0
Media storage	Supabase Free	1 GB file storage	₹0
Media processing	FFmpeg	Open source	₹0
Backend deployment	Render Free	Free tier, limitations apply	₹0
Frontend deployment	Render Free		₹0
Domain	Optional		₹0
Total fixed MVP cost			₹0
Actual variable cost	Mainly GPT-6 Astra	Usage dependent	Depends on usage